

**Math Intermediate Micro - Problem Set 3**  
**Due: Thursday, April 2**

**1. Cost Functions and Monopoly**

- (a) A firm produces output according to a production technology  $q = 1 + \sqrt{L}$  (where  $q$  is the amount produced of the firm's product, and  $L$  is its number of labor units hired). Find the firm's cost of producing  $q$  units,  $C(q)$ , if each labor unit costs \$2.
- (b) Suppose the firm is a monopolist, and faces demand curve  $Q^D(P) = 200 - 5P$ . How much will he sell, and at what price?

**2. Supply and Equilibrium.** Consider a market with demand curve  $Q = 200 - 5P$ . Firms are price-takers, and when there are  $N$  firms, each firm has cost function  $C(q) = f(N) + \alpha q^2$  (where  $f$  is a nonnegative function and  $\alpha > 0$ ).

- (a) Find the short-run equilibrium price with  $N = 10$  firms.
- (b) Find each firm's quantity, and the price, in a long-run equilibrium.
- (c) Write out the equation describing the long-run equilibrium # firms, and find it explicitly if  $\alpha = \frac{1}{5}$  and  $f(N) = N + 2$ .

**3. Double Auctions.** Assume that students 1-24 play in a double auction experiment. Odd students sell taxi rides to the airport. Even students need taxi rides to the airport. The odd students are willing to drive for an amount equal to two times their student number. The even students are willing to pay three times their student number for a ride.

- (a) Find the market clearing prices and quantities.
- (b) Now suppose the government imposes a tax \$ $t$  on buyers. What is the deadweight loss of a tax  $t = 2$ ? How many taxi rides will be exchanged, and what are the possible equilibrium prices?

**4. Double Auctions.** Consider a (continuous-no integer restriction in this one!) double auction with a set of buyers and sellers indexed by  $b$  and  $s$ , respectively, both distributed uniformly on the interval  $[0, 1]$ . A buyer  $b$  has valuation  $v(b) = b^{\frac{3}{2}}$  and is willing to buy  $d(b) = \frac{1}{2}b^2$  units. A seller  $s$  has cost  $c(s) = s$  and has  $q(s) = s$  units to sell. Find the market clearing price and quantity.